

Andrew Arochukwu

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EDUCATION

University of Waterloo

Bachelor of Honours Computational Mathematics

Waterloo, ON

January 2022 – May 2027

EXPERIENCE

Founding Software Developer

June 2025 – June 2026

Eign

Waterloo, ON

- Built the company's core geospatial analytics platform, owning the frontend, backend, authentication, API infrastructure, developer tooling, Stripe billing, and developer dashboards supporting customer onboarding and self-service platform management, from MVP through production.
- Engineered backend services integrating Databricks-generated datasets into production APIs, serving proprietary education, health, community, lifestyle, nature, future, and property value indices.
- Architected and deployed an embeddable, multi-tenant widget SDK (React + OpenStreetMap) utilizing custom geospatial visualization components to provide real-time location intelligence across third-party real estate platforms.
- Built a secure API key management infrastructure and control plane featuring client isolation, rate limiting, and usage tracking to support both widget distribution and standalone commercial index and POI/transit data APIs.
- Created an internal "POI Studio" visualization tool to audit data labeling and schema correctness, accelerating frontend widget feature development cycles for the engineering team.

Apache Airflow Contributor (RBC)

January 2025 – April 2025

RBC, MLH

Waterloo, ON

- Contributed production features and cross-platform regressions fixes to the Apache Airflow 3.0 beta core repository during release validation.
- Resolved a critical macOS-specific regression by replacing and restyling the core React code-viewing component, restoring developer workflow uniformity across operating systems.
- Engineered native copy-to-clipboard and source-code download actions within the Airflow UI, streamlining how developers interact with and share pipeline configurations.
- Designed and implemented a task instance log download feature integrated directly into Airflow's observability layer, enabling engineers to instantly extract local diagnostics for production failures.

Embedded Software Developer Intern

January 2024 – April 2024

Ciena

Ottawa, ON

- Built automated OSPF/ISIS regression testing framework for carrier-grade networking systems, reducing execution time by about 95% (2 hours to 5 minutes).
- Developed a JSON-driven device provisioning framework to replace legacy TL1 tooling, accelerating feature validation cycles for 15+ engineers and reducing simulator configuration times by about 90% (from 10 min to about 1 minute).
- Increased CPU/memory regression coverage by 40% by building performance data parsers for 6500-series network devices.

Software Developer Intern

January 2023 – August 2023

Ericsson

Ottawa, ON

- Built full-stack test analytics platform (React + Flask) used by 20+ engineers for real-time monitoring of distributed test pipelines
- Architected a migration from Excel-based test tracking to a relational PostgreSQL database, reducing query latency from 5s to less than 1s via an indexed schema redesign.
- Automated Grafana dashboard lifecycle management using Python, reducing configuration overhead and improving monitoring reliability.

TECHNICAL SKILLS

Languages: Python, TypeScript, Javascript, Go, Rust, C++, C, HTML/CSS

Frameworks/Libraries: React, Node.js, Next.js, React Native, FastAPI, Flask, PyTest, scikit-learn, Jinja

Tools & Platforms: Grafana, Prometheus, MySQL, Airflow, Docker, Nginx, PostgreSQL, Git